

Take-Home Challenge: Multi-Class Object Detection and Tracking in Highway Driving

Objective:

You are provided with a **front-facing dashcam video** (**challenge-m1e2.mp4**) of a vehicle driving on a highway. Your task is to detect and **track all nearby vehicles and pedestrians** across the entire video, assigning consistent IDs to each object over time.

This is a **multi-object tracking (MOT)** challenge using monocular video. You are expected to maintain tracking IDs across occlusions, exits, and re-entries — not just bounding boxes.

Input:

- **challenge.mp4**: A highway-driving video from a monocular front-facing camera
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Submission Requirements:

1. **Output video (output_tracking.mp4)**:
 - Each detected object must be bounded with a box
 - Class label (**Car**, **Truck**, **Motorcycle**, **Bus**, **Pedestrian**) and unique track ID must be rendered over each box
 - Objects should **retain the same ID across frames** unless they leave and re-enter the scene
 2. **README (markdown or PDF)**:
 - Models used for detection
 - Tracking methodology
 - Performance optimization steps
 - How you handle occlusion, entry/exit of objects, frame drops
 - Any assumptions made
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You May Use:

- Any object detector
- Any MOT library or your own implementation
- Tools: OpenCV, PyTorch, NumPy, ONNX

Evaluation Criteria:

- **Tracking accuracy** (visual ID consistency over time)
- **Detection precision** (avoid false positives)
- **Robustness** to occlusion, sudden motion, lighting variation
- **Code structure and clarity**

Bonus Points:

- Predict approximate velocities for each vehicle along with tracking information
- Use motion models like Kalman filters or optical flow refinement
- Output a CSV: `[frame_id, object_id, class, bbox]`

[challenge-mle2.mp4](#)